

Sensory receptors and coding

The written notes for the sensory receptors and coding walkthrough. Week 5 competencies 8 to 12.

Transduction

1. A stimulus opens channels in the receptor membrane (for touch, mechanically gated cation channels).
2. Sodium enters and the ending depolarizes: a receptor potential. It is a graded potential: bigger for a stronger stimulus, fading as it spreads.
3. If the receptor potential is still above threshold at the trigger zone, the trigger zone fires action potentials.
4. The action potentials are all the same size and travel unchanged along the axon. At the trigger zone, information carried by size becomes information carried by frequency.

Sensory transduction

Converting the energy of a stimulus into a change in membrane potential.

Receptor potential

The graded potential produced in a sensory receptor by a stimulus.

Frequency coding

A stronger stimulus produces a larger receptor potential and a higher firing frequency.

Population coding

A stronger stimulus also reaches the thresholds of more receptors, so more sensory neurons fire.

Receptor classes

Class	Stimulus	Example locations
Mechanoreceptor	Pressure, stretch, vibration, sound	Skin, muscle spindles, baroreceptors in the carotid arteries, hair cells of the ear
Chemoreceptor	Particular molecules	Carotid body (blood O ₂ , CO ₂ , pH), taste buds, olfactory epithelium
Thermoreceptor	Temperature	Skin, hypothalamus
Photoreceptor	Light	Retina (rods and cones)
Nociceptor	Stimuli strong enough to damage tissue	Skin, joints, muscles, internal organs

Most receptors are inside the body, reporting on conditions you are not aware of.

What the brain is told

Property	How it is coded
Modality (which sense)	Labeled line: which pathway is active and where it ends in the cortex
Location	Which sensory neurons fire, set by their receptive fields; sharpened by lateral inhibition
Intensity	Firing frequency, and the number of receptors firing
Duration	How long the firing lasts, which depends on adaptation

- Labeled line: pressure on a closed eye is seen as light, because the signal arrives in the visual cortex.
- Law of projection: stimulating a sensory nerve anywhere along its length produces a sensation that seems to come

from the receptors at its end. Phantom limb pain is an example.

Receptive fields and acuity

- A receptive field is the area that changes one sensory neuron's firing when stimulated.
- Small, dense receptive fields (fingertips, lips) give fine acuity and two-point thresholds of a few millimeters. Large, sparse fields (the back) give thresholds of several centimeters.
- Two points are felt as two when they stimulate different receptive fields with an unstimulated field between them.
- Lateral inhibition: a strongly stimulated neuron inhibits its neighbors through interneurons, increasing the contrast between center and edge so the stimulus is located more precisely.

Adaptation

	Tonic (slowly adapting)	Phasic (rapidly adapting)
Response to a steady stimulus	Keeps firing, slowing a little	Fires at onset, falls silent, often fires again at offset
Reports	The stimulus, continuously	Change
Examples	Muscle spindles, baroreceptors, nociceptors	Pacinian corpuscles (vibration), hair follicle receptors

Each receptor's adaptation matches its job. Monitoring blood pressure with a phasic receptor would leave a sustained drop unreported after its first moment.